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unun-vna 0.7.1

unun-vna is a command-line tool for measuring the **small-signal efficiency** of HF UnUn transformers with an S-A-A-2 / NanoVNA V2 using the **one-transformer method**.

The tool is generalized:

- no nominal load/series resistor is assumed;
- the actual series-load fixture is measured as a complex impedance versus frequency;
- transformer efficiency is calculated from measured S11/S21 and the measured fixture;
- transformer impedance ratio is configurable;
- 1:49 remains the default only for convenience;
- analyzed data can be plotted directly as VSWR and efficiency curves.

The lower-impedance side of the transformer is connected to Port 1. The higher-impedance side is connected to the series fixture and then Port 2.

1. Transformer ratio

`--ratio` means the **impedance** ratio

$$N = \frac{Z_{\text{high}}}{Z_{\text{low}}}$$

These are equivalent:

```
--ratio 49
--ratio 1:49
--ratio 1/49
```

All mean N = 49.

Examples:

```
1:4   -> N=4,  turns ratio 1:2
1:9   -> N=9,  turns ratio 1:3
1:16  -> N=16, turns ratio 1:4
1:49  -> N=49, turns ratio 1:7
1:64  -> N=64, turns ratio 1:8
```

The default is N=49.

Ratios below 1 are rejected because this measurement topology expects the low-impedance side at Port 1. Reverse the transformer connection instead.

2. Generic measurement topology

```
Port 1 ---- UnUn ---- measured series fixture ---- Port 2
      Z0      1:N                      Z0
```

Z0 is normally 50 Ω .

The fixture may have any resistance value. It is not inferred from the transformer ratio and it is not assumed to equal its nominal/DC value.

It is measured as

$$Z_s(f) = R_s(f) + jX_s(f).$$

3. Ideal matching load for a selected ratio

For an ideal transformer with ratio N, the target total high-side resistance for a perfect resistive match is

$$R_{\text{high,target}} = NZ_0.$$

Port 2 itself contributes Z0, so the ideal purely resistive series fixture would be

$$R_{\text{series,target}} = (N - 1)Z_0.$$

At Z0=50 Ω :

UnUn	N	Total high-side target	Ideal series fixture
1:4	4	200 Ω	150 Ω
1:9	9	450 Ω	400 Ω
1:16	16	800 Ω	750 Ω
1:25	25	1250 Ω	1200 Ω
1:49	49	2450 Ω	2400 Ω
1:64	64	3200 Ω	3150 Ω

These numbers are **diagnostics only**. They are never substituted for the measured fixture impedance.

4. Fixture characterization

For a pure series impedance between two equal Z_0 ports:

$$S_{11} = \frac{Z_s}{2Z_0 + Z_s}$$

$$S_{21} = \frac{2Z_0}{2Z_0 + Z_s}.$$

Therefore:

$$Z_{s,1} = 2Z_0 \frac{S_{11}}{S_{21}}$$

and

$$Z_{s,2} = 2Z_0 \left(\frac{1}{S_{21}} - 1 \right).$$

A pure series element also satisfies

$$S_{11} + S_{21} = 1.$$

The **fixture** command reports both impedance estimates, their disagreement, and **|S11+S21-1|**.

This checks whether the physical fixture can safely be represented by one complex series impedance.

5. Efficiency does not depend on the configured ratio

By S-parameter definition,

$$|S_{21}|^2$$

is the incident Port-1 power fraction delivered to the matched Port-2 termination Z_0 .

The same current flows through the Port-2 resistance and the series fixture. Thus total real high-side load power is

$$\frac{P_{\text{high}}}{P_{\text{incident}}} = |S_{21}|^2 \left(1 + \frac{\Re\{Z_s\}}{Z_0} \right).$$

Power accepted at Port 1 is

$$\frac{P_{\text{accepted}}}{P_{\text{incident}}} = 1 - |S_{11}|^2.$$

Therefore

$$\boxed{\eta = \frac{|S_{21}|^2 \left(1 + \frac{\Re\{Z_s\}}{Z_0} \right)}{1 - |S_{11}|^2}}$$

and

$$L_{\text{dB}} = -10 \log_{10}(\eta).$$

VSWR

The analysis also reports Port-1 VSWR derived directly from calibrated S11. With

$$\rho = |S_{11}|,$$

$$\boxed{\text{VSWR} = \frac{1 + \rho}{1 - \rho}}$$

for $\rho < 1$.

A perfect match gives $\text{VSWR} = 1.0$. Total reflection gives infinite VSWR. Values with $|S_{11}| \geq 1$, which can arise from residual calibration or measurement error, are reported as infinite rather than as a non-physical negative VSWR.

The tool also reports the **ideal lossless VSWR** for the configured transformer ratio with the actually measured fixture. This helps separate load mismatch from transformer loss.

There is no N in the measured efficiency equation. That is intentional.

The measured power flow already contains the transformer's actual behaviour; the ratio is not needed to reconstruct efficiency.

6. What --ratio is used for

The configured ratio is used to calculate a **lossless ideal reference** for the actual measured fixture.

The real measured high-side load is

$$Z_{\text{load}} = Z_s + Z_0.$$

An ideal lossless transformer with impedance ratio N reflects that to Port 1 as

$$Z_{\text{in}} = \frac{Z_{\text{load}}}{N}.$$

The input reflection expected from load mismatch alone is

$$\Gamma_{\text{ideal}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}.$$

Therefore

$$|S_{11,\text{ideal}}| = |\Gamma_{\text{ideal}}|.$$

The accepted incident power of that ideal network is

$$1 - |\Gamma_{\text{ideal}}|^2.$$

Since an ideal transformer is lossless, the fraction of that real power dissipated in Port 2 is

$$\frac{Z_0}{Z_0 + \Re\{Z_s\}}.$$

Hence

$$|S_{21,\text{ideal}}|^2 = (1 - |\Gamma_{\text{ideal}}|^2) \frac{Z_0}{Z_0 + \Re\{Z_s\}}.$$

The analysis output shows measured S11/S21 beside these ideal values.

This is useful when the fixture is not exactly the perfect matching resistance for the selected ratio.

7. S-A-A-2 enhanced-response calibration

The host-side S21 calibration follows the NanoVNA V2 **enhanced-response** user-calibration algorithm used for the on-screen traces. This is important for the one-transformer method because the series-load fixture is normally highly reflective.

7.1 Reflection calibration

Port-1 S11 uses the normal three-term SOL error model

$$m = e_{00} + \frac{e_t \Gamma}{1 - e_{11} \Gamma},$$

where e_{00} is directivity, e_t is reflection tracking and e_{11} is source match. SHORT, OPEN and LOAD determine these terms.

7.2 Transmission leakage

The NanoVNA V2 firmware models residual forward leakage as an affine function of the raw reflection measurement:

$$L(m_{11}) = L_0 + L_r m_{11}.$$

The two leakage states are taken from **SHORT and OPEN S21**. Both standards ideally have zero true transmission, so their measured S21 is treated as feed-through/leakage.

7.3 THRU reflection is part of the calibration

The THRU capture now stores both raw S11 and raw S21. DUT and THRU leakage are corrected with their own reflection states:

$$T_{\text{DUT}} = m_{21,\text{DUT}} - L(m_{11,\text{DUT}}),$$

$$T_{\text{THRU}} = m_{21,\text{THRU}} - L(m_{11,\text{THRU}}).$$

The THRU is therefore not assumed to have exactly zero reflection.

7.4 Source-match loop-gain correction

After SOL calibration, the NanoVNA V2 enhanced-response algorithm calculates

$$G(\Gamma) = \frac{1}{1 - e_{11}\Gamma}.$$

For the DUT and THRU this gives

$$G_{\text{DUT}} = \frac{1}{1 - e_{11}S_{11,\text{DUT}}},$$
$$G_{\text{THRU}} = \frac{1}{1 - e_{11}S_{11,\text{THRU}}}.$$

The calibrated forward transmission is then

$$S_{21} = \frac{T_{\text{DUT}}}{T_{\text{THRU}} G_{\text{DUT}}/G_{\text{THRU}}}$$

or equivalently

$$S_{21} = \frac{T_{\text{DUT}}}{T_{\text{THRU}}} \frac{1 - e_{11}S_{11,\text{DUT}}}{1 - e_{11}S_{11,\text{THRU}}}.$$

For a well-matched DUT this correction is small. For a fixture with $|S_{11}|$ close to one it can be large enough to matter by more than a dB.

8. S-A-A-2 Port-2 load-match limitation

The S-A-A-2 is a T/R VNA, not a full bidirectional two-port VNA.

It measures S11 and forward S21, but cannot perform a complete independent Port-2 load-match correction.

High-ratio transformer/fixture combinations can be strongly reflecting when viewed from Port 2 even when Port 1 is well matched.

A THRU verification can therefore look excellent while a highly reflective DUT still has biased S21.

Recommended Port-2 pad

Use a fixed 50 Ω attenuator, typically 10 dB, immediately before Port 2:

Port 2 ---- 10 dB pad ---- calibration plane ---- DUT

The same pad must remain in place during:

1. calibration;
2. fixture characterization;
3. transformer measurement.

The THRU calibration removes its forward loss and phase. Reflections from Port 2 pass through it twice, so a 10 dB pad reduces the load-match influence by roughly 20 dB.

9. Calibration

```
unun-vna calibrate \  
  --start 1M \  
  --stop 35M \  
  --points 501 \  
  -o hf35-pad
```

The program requests:

1. SHORT on Port 1 - both S11 and S21 are retained;
2. OPEN on Port 1 - both S11 and S21 are retained;
3. 50 Ω LOAD on Port 1 - used for the SOL reflection model;
4. THRU between the calibration planes - both S11 and S21 are retained.

SHORT/OPEN S21 define the leakage model. THRU S11 is required for the source-match enhanced-response correction.

Files:

```
hf35-pad.npz  
hf35-pad.npz.json  
hf35-pad.npz.through.csv
```

10. Characterize a fixture

The fixture extraction itself is independent of ratio.

For a 1:9 reference:

```
unun-vna fixture \  
  --cal hf35-pad.npz \  
  --ratio 1:9 \  
  -o fixture-1to9
```

For 1:64:

```
unun-vna fixture \  
  --cal hf35-pad.npz \  
  --ratio 64 \  
  -o fixture-1to64
```

--ratio changes only the target-load diagnostics printed by this command.

11. Measure a transformer

Measurement acquisition itself requires no ratio:

```
unun-vna measure \  
  --cal hf35-pad.npz \  
  -o transformer
```

The measured chain is:

Port 1 ---- transformer ---- characterized fixture ---- Port 2 pad ---- Port 2

12. Analyze

Default 1:49:

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture.npz \  
  -o efficiency.csv
```

Create the analysis CSV and a plot in one step:

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture.npz \  
  -o efficiency.csv \  
  --plot efficiency.svg
```

Explicit 1:49:

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture.npz \  
  --ratio 1:49 \  
  -o efficiency.csv
```

1:9:

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture-1to9.npz \  
  --ratio 1:9 \  
  -o efficiency-1to9.csv
```

1:64:

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture-1to64.npz \  
  --ratio 64 \  
  -o efficiency-1to64.csv
```

13. Analysis CSV

The analysis CSV contains:

```
frequency_hz  
s11_db  
vswr  
s21_db  
fixture_r_ohm  
fixture_x_ohm  
impedance_ratio_high_over_low  
target_high_side_resistance_ohm  
target_series_resistance_ohm  
ideal_lossless_s11_db_for_measured_fixture  
ideal_lossless_vswr_for_measured_fixture  
ideal_lossless_s21_db_for_measured_fixture  
accepted_fraction  
delivered_fraction  
efficiency
```

efficiency_percent
transformer_loss_db

The `target_*` and `ideal_lossless_*` columns depend on `--ratio`.

The actual efficiency columns do not.

14. Visualization

`unun-vna plot` turns one or more analysis CSV files into a simple two-panel line graph intended for practical amateur-radio evaluation:

1. **VSWR** versus frequency;
2. **transformer efficiency** in percent versus frequency.

Both panels share the same frequency axis. By default the 40 m, 20 m, 15 m and 10 m IARU Region-1 amateur allocations are shaded so that the useful parts of a wide 1.35 MHz sweep are immediately visible. The marked ranges are 7.000-7.200, 14.000-14.350, 21.000-21.450 and 28.000-29.700 MHz.

For a single transformer:

```
unun-vna plot efficiency.csv -o efficiency.svg
```

The output format is selected from the filename extension:

```
.png    raster image
.svg    vector image, useful for README/GitHub
.pdf    vector PDF
```

The default VSWR display range is 1.0 to 3.0. Higher measured values are not modified in the CSV; they simply lie above the visible plot range. Override the display limit when required:

```
unun-vna plot efficiency.csv --vswr-max 5 -o efficiency.png
```

The lower efficiency-axis limit is chosen automatically by rounding down to a 10-percentage-point boundary while retaining the complete data range. It can also be set explicitly:

```
unun-vna plot efficiency.csv --efficiency-min 50 -o efficiency.svg
```

Compare multiple transformers

Several analysis files can be plotted together. The same series ordering is used in both panels so that each transformer has the same line color in the VSWR and efficiency graphs:

```
unun-vna plot \  
  unun1-efficiency.csv \  
  unun2-efficiency.csv \  
  --label "Tubular autotransformer" \  
  --label "Toroid 7+7 crossover" \  
  --title "1:49 UnUn comparison" \  
  -o comparison.svg
```

If no `--label` options are given, labels are derived from the CSV filenames. When labels are supplied, provide exactly one per input file.

Band shading can be disabled with:

```
unun-vna plot efficiency.csv --no-bands -o efficiency.svg
```

The plotter reads only `frequency_hz`, `vswr` and `efficiency_percent` from the analysis CSV. It performs no RF calculations, so visualization cannot alter the measured efficiency result.

15. Ideal sanity checks

1:9

At $Z_0=50\ \Omega$, use an ideal $400\ \Omega$ series fixture:

$$400 + 50 = 450\ \Omega$$

and

$$450/9 = 50\ \Omega.$$

Thus an ideal transformer has

$$S_{11} = 0$$

and

$$|S_{21}| = 1/3$$

or approximately

$$S_{21} = -9.54\ \text{dB}.$$

Efficiency evaluates to 100%.

1:49

With $2400\ \Omega$ series:

$$|S_{21}| = 1/7$$

or

$$S_{21} \approx -16.90\ \text{dB}.$$

1:64

With $3150\ \Omega$ series:

$$|S_{21}| = 1/8$$

or

$$S_{21} \approx -18.06\ \text{dB}.$$

16. Remaining assumptions

The tool does **not** assume:

- a 1:49 transformer;
- a 2400 Ω fixture;
- any nominal fixture resistance;
- that the fixture is purely resistive.

It does assume:

1. the lower-impedance transformer side is connected to Port 1;
2. the fixture is adequately represented by one complex series impedance;
3. Port 2 is a matched real Z0 termination at the calibrated reference plane;
4. calibration, fixture measurement and DUT measurement use the same RF setup;
5. a fixed Port-2 pad, if used, remains in place;
6. the measurement is small-signal.

A fully arbitrary two-port fixture would require S12 and S22 and therefore a full bidirectional VNA.

17. Full workflow example

```
pipx install --editable . --force
```

```
unun-vna info
```

```
unun-vna calibrate \  
  --start 1M \  
  --stop 35M \  
  --points 501 \  
  -o hf35-pad
```

```
unun-vna fixture \  
  --cal hf35-pad.npz \  
  --ratio 1:49 \  
  -o fixture
```

```
unun-vna measure \  
  --cal hf35-pad.npz \  
  -o transformer
```

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture.npz \  
  --ratio 1:49 \  
  -o efficiency.csv \  
  --plot efficiency.svg
```

Or compare several previously analyzed transformers

```
unun-vna plot unun1.csv unun2.csv -o comparison.svg
```

18. Source structure

```
unun-vna/
|-- pyproject.toml
|-- README.md
|-- src/
|   |-- unun_vna/
|       |-- __init__.py
|       |-- cli.py
|       |-- analysis.py
|       |-- plotting.py
|-- tests/
    |-- test_math.py
    |-- test_calibration.py
    |-- test_plotting.py
```

`analysis.py` deliberately separates:

- ratio-independent measured efficiency;
- ratio-dependent ideal-reference diagnostics.

`plotting.py` contains only CSV reading and visualization. It does not duplicate or modify the RF calculations.

This prevents a nominal transformer ratio or a visualization option from silently changing the measured power calculation.